Expected = 63

22516

12425

03 Hours / 70 Marks Seat No. 1 4 8 8 8 4

Instructions – (1) All Questions are *Compulsory.*

(2) Illustrate your answers with neat sketches wherever necessary.

(3) Figures to the right indicate full marks.

(4) Assume suitable data, if necessary.

(5) Mobile Phone, Pager and any other Electronic Communication devices are not permissible in Examination Hall.

Marks

1. Attempt any FIVE of the following: **10**

a) State any two command line based operating systems and two GUI based operating system.

10

b) Define system call and enlist its types.

c) Define PCB with suitable diagram.

d) Describe CPU and I/O burst cycle with suitable diagram.

e) State any four criteria in CPU scheduling.

f) Define the term paging.

g) Draw the structure of unix operating system.

P.T.O.

Marks

2. **Attempt any THREE of the following:** 12

a) Differentiate between multitasking and multiprogramming.

4 b) List activities performed in process management and file management.

11

3 c) Define the term inter process communication. Explain any one technique of IPC.

4 d) Describe necessary conditions leading to deadlock.

3. **Attempt any THREE of the following:** 12

3 a) Difference between user level and kernel level thread. (Any four points)

4 b) Describe First Come First Served (FCFS) scheduling algorithm with example.

11

c) Describe the concept of segmentation in Operating System with suitable example.

4 d) Describe Single level and two level directory structure.

4. **Attempt any THREE of the following:** 12

4 a) Describe multiprocessor system with advantage and disadvantage.

10

4 b) List and describe any four services of operating system.

c) Describe the concept of scheduling queues with diagram.

2 d) Write steps of Banker's algorithm to avoid deadlock.

e) Explain swapping in operating system with diagram and example.

Marks

5. Attempt any TWO of the following: 12

a) Describe use of following commands with syntax:

5

i) ps

ii) wait

iii) sleep

iv) kill

b) Write two uses of following operating system tools:

11

i) User management

ii) Security Policy

iii) Performance Monitor

c) Given a page reference string with (03) pages frames. Calculate the page faults with "Optimal" and "LRU" page replacement algorithm respectively.

6

7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7.

6. Attempt any TWO of the following: 12

a) Calculate average waiting time for SJF (Shortest Job First) and Round Robin (RR) algorithm table: (Time slice 4 ms)

6

Process	Burst Time
p1	10
p2	4
p3	9
p4	6

10

b) Explain linked and indexed file allocation method with neat diagram.

4

c) Describe variable partitioning technique of memory management with suitable example.

B1

PY-159

f

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2324

(MAIN ANSWER BOOK)

To be filled by the Candidate

Subject :- Operating Systems(OSY)

Subject Code :- 2 2 5 7 6

Section :- I / II

Course and Year/Sem :- CO - 5I

Date of Exam:- 0 3 N 2 2 0 2 4

Main Ans. Book	No. of Supplements	Total
1	01	02

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Sr. No. 29692946

Examination Seat No. 884

1	4	8	8	8	4

Exam. Center Code	Institute Code
1602	160[illegible]

Signature of block Supervisor / Invigilator

[signature]

Q. No.	a	b	c	d	e	f	g	h	i	j	k	l	m	n	Total	Q.C.
1		2	2	2	2	2									10	10
2		04	04	04											12	12
3	04	04		04											12	12
4	04	04		03											11	11
5	06		06												12	12
6	06		06												12	12
															69	69

MSBTE PHOTOCOPY AB

Q. No	UV/OV/PH Marks: Verifier Marks	UV/OV/PH Marks: Marks Validation by 3 member Committee
1	10	
2	12	
3	12	
4	11	
5	12	
6	12	
7		
8		
Total	69	
Sign	[signature]	

Marks Validation by RBTE 3 member Committee

Marks on Answer book before Verification	Marks on Answer book after Verification

Valid Change in Verification marks

From ______ to ______

Change / No Change

Dated Signatures

1)

2)

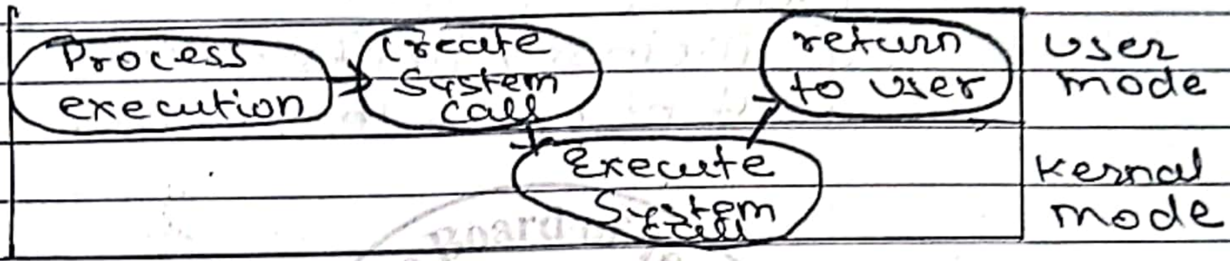
3)

Q1

b) System call

System call is call that provide OS services to user via API (Application Programming Interface).

It is the interface between user and kernal.



Following are types of system call :-

1) Process Control
2) File management
3) Device management
4) Communication
5) Information maintenance.

c) PCB

PCB stands for Process Control Block.

It is a Data Structure which stores all the information related to process.

Information like :- pointer, state of process, counter, register etc...

It is also known as Transfer Control Block (TCB).

P.T.O

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Diagram

Pointer	Process State
Process Number (PID)	
Program counter	
CPU Registers or Registers	
Memory limits	
List of open files	
.....	

02

fig: PCB

d) CPU burst cycle

Time peroid process is busy with the CPU (execution) is known as CPU burst Cycle.

I/O burst Cycle

Time peroid process is busy with the I/O devices (such as printer, scanner, peripheral devices) known as I/O burst cycle.

P. T. O

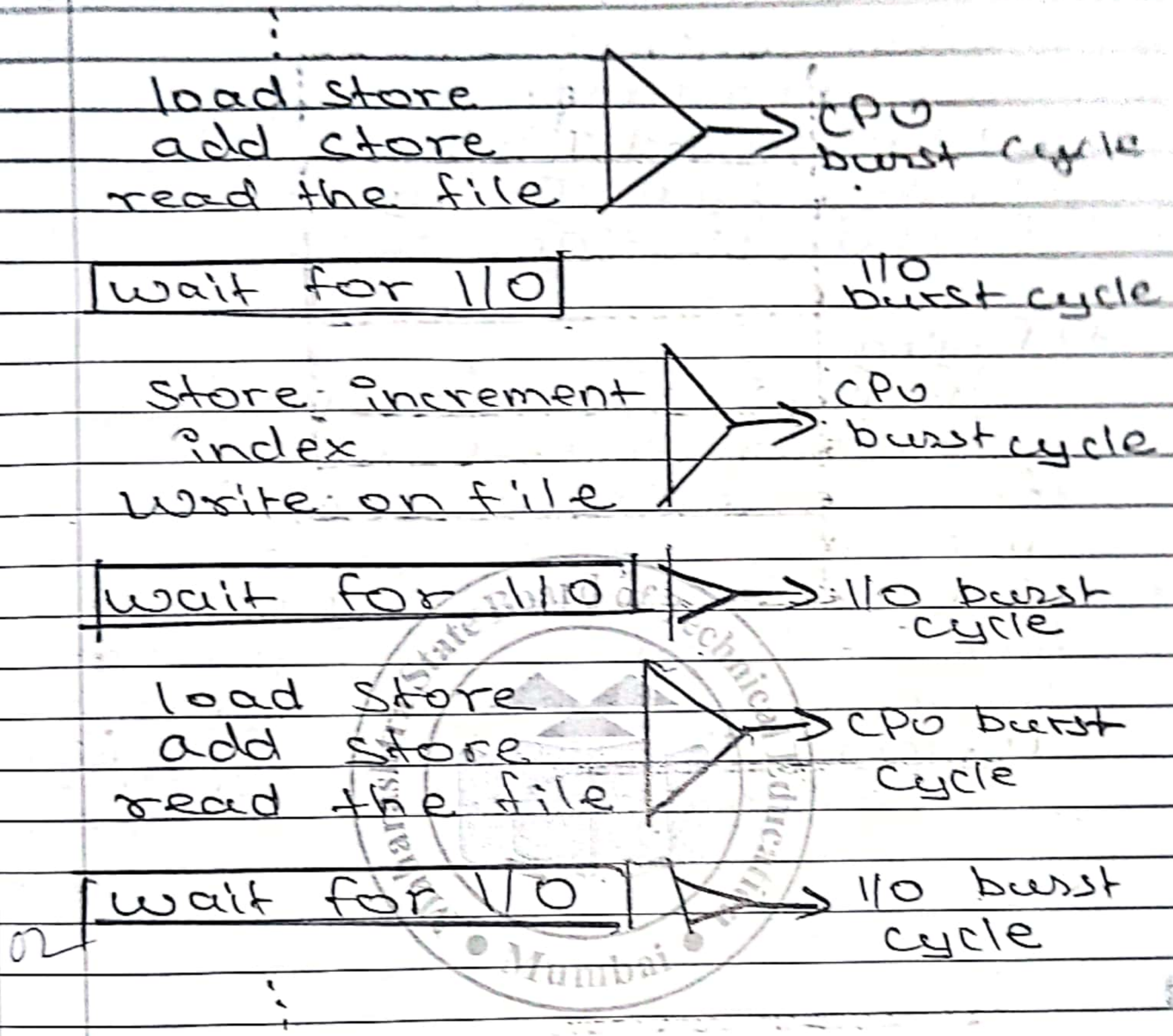

fig: Diagram of I/O burst and CPU burst Cycle

e) Following are the scheduling Criteria:

1. Response Time
2. Waiting Time
3. Turnaround Time
4. ~~Throughghput~~ Througput
5. CPU utilization

f) **Paging**

Paging is a storage mechanism which ~~retri~~ retrieves the process from Secondary memory to main-memory as pages.

Process of ~~retriving~~ ~~pa~~ memory ~~from of~~ in form of pages from Secondary memory to main Memory is called as Paging.

Paging is fixed in size.

Q2.

d) Deadlock

Deadlock is a situation in which two or more processes are unable to proceed due to resources are hold by other process.

This results in infinite waiting state of process.

Following are necessary conditions due to that deadlock occur.

1) Mutual Exclusion

Resources of each process or one of the process must be in non-sharable mode.

∴ Only one process can share resources at a time.

2) Hold and wait

The process hold one or more resources and wait for additional resources hold by the other processes

3) No - Preemption

Process can't take resources forcibly from the other process.

They must be released voluntarily.

P. T. O

4) <u>Circular wait</u>

There is circular chain of process, where each process needs the resources hold by other process in chain.

Example, P_1, P_2, P_3 are process and R_1, R_2, R_3 are resources

P_1 Needs Resources hold by P_2
P_2 Needs Resources hold by P_3
04 P_3 Needs Resources hold by P_1

c) <u>Inter-process communication</u>

when processes share data among themselves data can be some kind of information, the communication which is established is known as Inter-process communication

Two co-operative process communicate with each other known as Inter-process communication.

There are two model or types of Inter-process communication.

1. Shared memory.
2. Message Passing.

Shared Memory

In this process create a specific memory region to communicate.

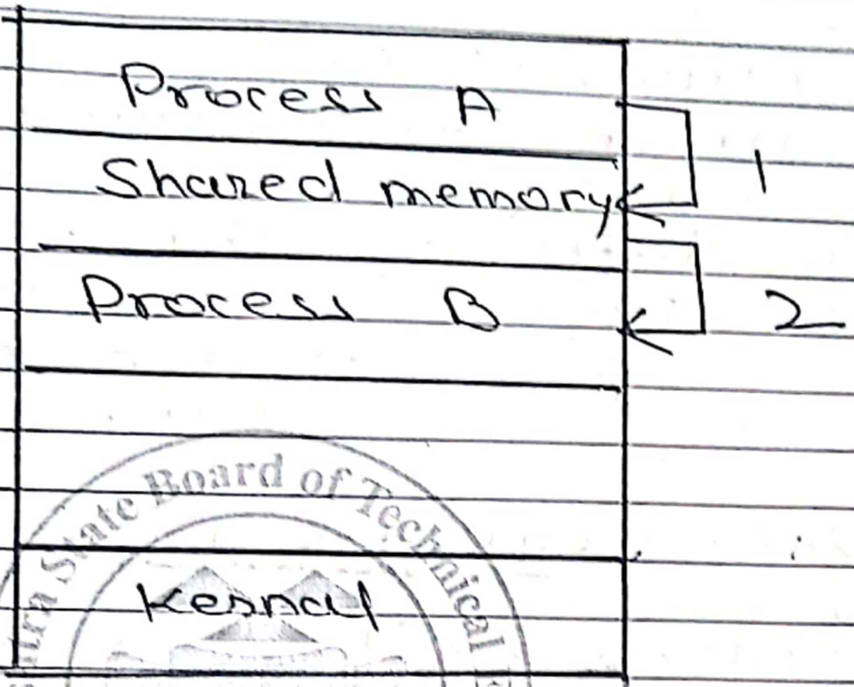

fig: Shared Memory

In this mechanism, both the Co-operative process share some
04 region which is termed as "shared memory" where they can write and read data.

In this region, information and data is ~~stor~~ shared but Kernal is ~~an~~ unaware about it.

Process exchange information Among themselves without use of O.S (Kernal) is Shared Memory

Advantages
① Simple to Implement
② Effective in networks.

DisAdvantages
① Can't share data or info. when processes resides in different systems.
② Faster Once but as compared to message passin security issue.

2. Message Passing

1) Message Passing is a message in which process communicates through help of message.
2) Message or data or passed with the help of kernel (O.S).
3) It is more effective when they or on other computer.
4) They are more secured than others.
5) ~~Fast~~ slower but they execute fastly.

P.T.O

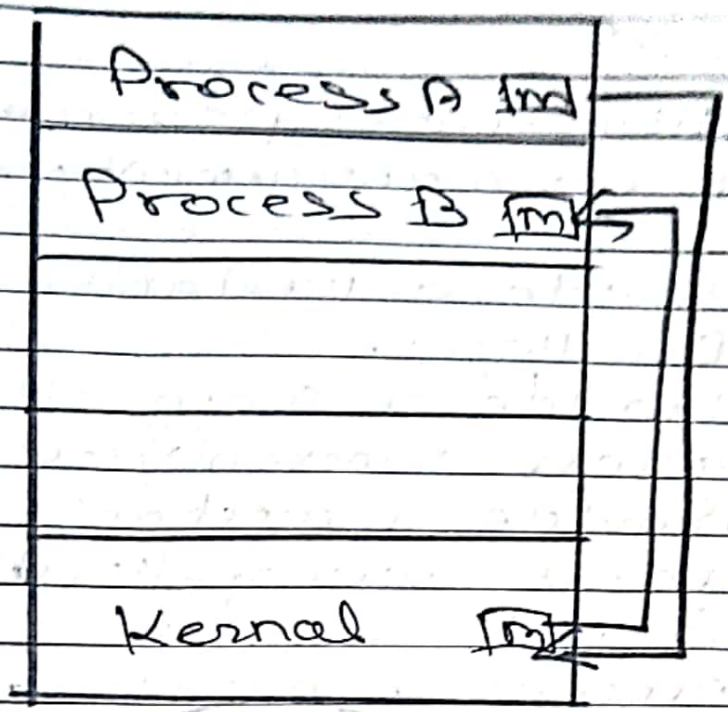

fig: Message Passing

Process use kernal as interface to communicate within threads.

For example, Process A can share message to kernal. and kernal share it to the Process A.

Advantage

- Most secured mechanism
- Slower - But Effective.

DisAdvantages

- Slower in speed.
- ~~here~~ there is intermidate message passing.

b) ~~Acitt~~ Activities performed by process management.

1. Provide a mechanism for Deadlock.
2. Provide a mechanism for process syncronization.
3. Provide a mechanism for process monopolization.
4. Provide a mechanism for process scheduling.
5. creating, deleting, termination process among system.
6. Divides processes into preemptive higher and lower priorites by premptive scheduling.
7. Defines various process states.

Activites performed by File management.

1. Free space management by techniques Bit map & Linked List.
2. Creating File
3. Deleting a file
4. Write on a file
5. Read a file
6. Rename a file
7. Truncating the file
8. Repositining a file
9. Accessing a file by - Random

P. T. O

sequential, Swapping method.
(D) Allocating files by - Linked list allocation, contigoues allocation index allocation.

04

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Q3.

d) Directory Structure

1) To organize file efficiently operating system provide a concept of Directory structure.
2) In this we can manage or group of file efficiently.

Three types
1) Single-level Directory structure
2) ~~twot~~ two-level structure
3) Tree-level structure.

Single-level structure

1) Single-level directory is simple and easiest to implement.
2) It contains files in single directories with multiple entries.
3) All file contains in same directory, Hence that directory is known as root Directory.
4) The files in directory can't have same name, you user have to name it uniquely.
5) In case, same name the name is overridden over ~~prir~~ priviously.

P.T.O

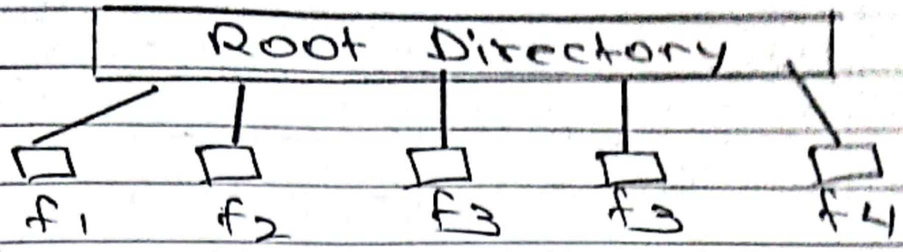
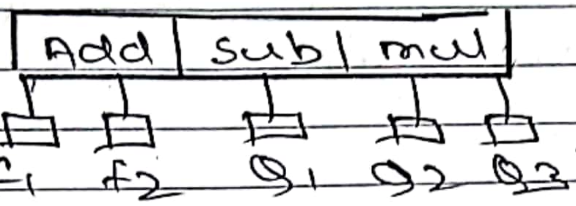


Fig: Singlelevel

example



Two-level Structure

1) Due to Drawbacks of single-level Directory structure there was similiar structure Two-level Directory structure.
2) Here there is root directory (MFD) master file directory divided into users.
3) User can have their own single-level directory (UFD) User File Directory.
4) If user wants to check any file or search/sort user needs to go through root directory.
5) Directories can't have unique name but users can differentiate the name of file give same file name.

P.T.O

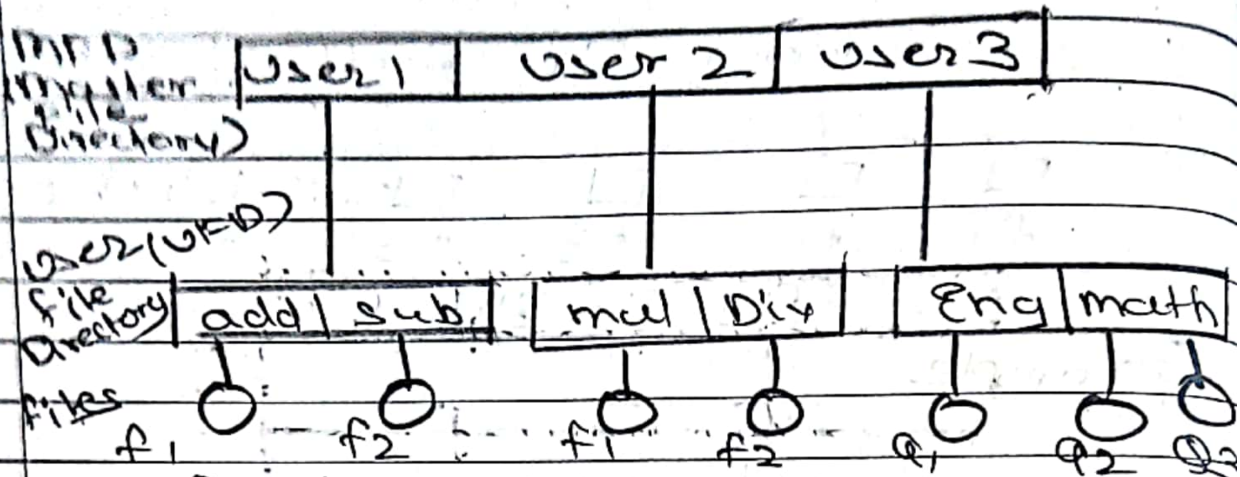


fig: two-level structure

Q)

b) First Come First Serve (FCFS)

1) It is an non-preemptive scheduling alogrithm which serve processes on queue and execute them.
2) Process which come first is allocate CPU first.
3) Real-life example, for ticket booking there is Queue we follow that Queue who comes first that gets ticket first
4) Same way, in scheduling which process comes first get execute and exitted first.
5) In multilevel Batch Queue, this scheduling algorithm is used for Background Queue.
6) Based on Arrival time means when the process has come based on that it execute.

P.T.O

Example: There are processes given with arrival and Burst time. calculate Average Turnaround and waiting time with FCFS.

Process	Arrival time	Burst Time
P_1	0	10
P_2	1	10
P_3	2	5

GANTT chart

P_1	P_2	P_3

0   10   20   25

Completion Time

$P_1 = 10$ $P_2 = 20$ $P_3 = 25$

Turnaround Time (C.T - A.T)

04 (Completion Time - Arrival time)

$P_1 = 10 - 0 = 10$

$P_2 = 20 - 1 = 19$

$P_3 = 25 - 2 = 23$

Average turnaround time

$$\frac{P_1 + P_2 + P_3}{3} = \frac{10 + 19 + 23}{3}$$

$= 17.33$ ms/unit

P.T.O

Waiting Time (TAT - B.T)
(Turnaround time - Burst time)

$P_1 \to 10 - 10 = 0$
$P_2 \to 19 - 10 = 9$
$P_3 \to 23 - 5 = 18$

Average waiting time

$$\frac{P_1 + P_2 + P_3}{3} = \frac{0 + 9 + 18}{3}$$

= 9 ms/unit

a)

Parameters	User Thread	Kernal Thread
Definition	The thread created by user in user space is known as user thread.	The thread created by kernal in kernal space is kernal thread
Involvement	There is not involvement of O.S	There is involvement of O.S
Execution	They are faster	They are slower
Creation	Creation is easier of user Thread	Creation is burden for kernal Thread

P.T.O

limitation	There are nor limitation in User thread by O.S	There are limitation in Kernel thread by O.S
Management	Management is Done in user space/mode	management is done in Kernel mode
System calls	System calls are not generated among user thread.	System calls are generated in Kernel Thread.
Works on	Application software	System O.S

04

Q4

a) Multiprocessor System

1) When more than one CPU are user in execution of the processes is known as multiprocessor system.
2) Many CPU are assigned in one cabinet to worked in one cabinet for closed communication.
3) Multiprocessor system are also known as Parallel Systems or tighed coupled systems.
4) It is illustrated as multi-processor which leads to better throughput rate.
5) Here CPU shares Registers and cached memory and connected main memory.

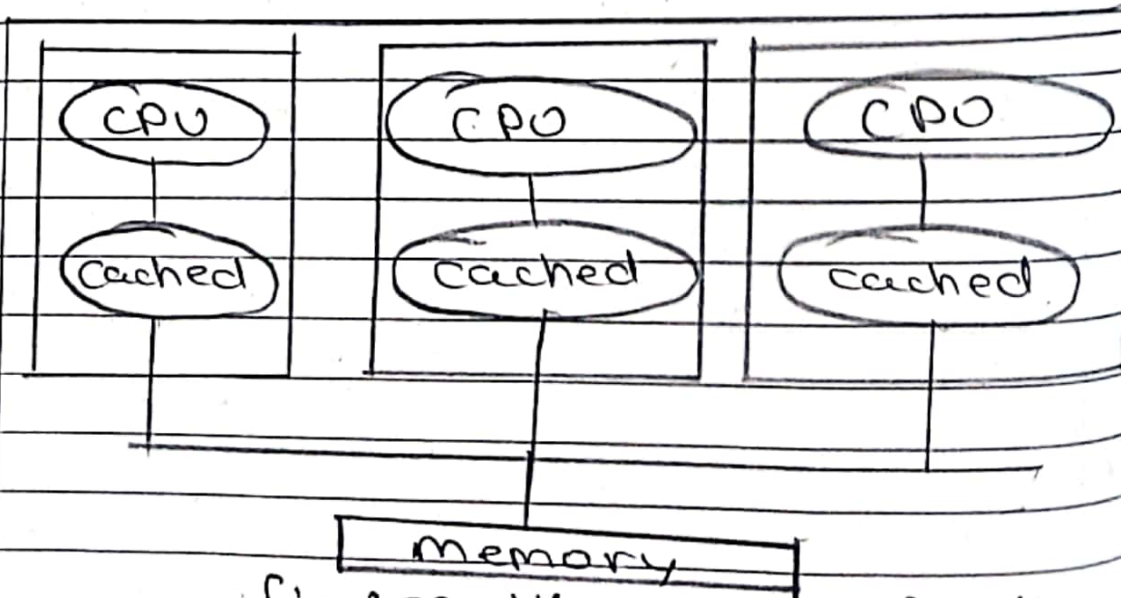


fig: multiprocessor System.

P.T.O

There are two types of multiprocessor System:
1. Symmetric System
2. ASymmetric System.

1. Symmetric System
There is single multiple CPU which are treated equally.
No Master-slave relationship.

2. ASymmetric System
There is one main CPU and other CPU are followed or instructed by main. Follows Master-slave Relationship.

* Advantages
1) Faster response as process execute faster, increases response time
2) Better throughput rate.
3) Parallel Systems: Processes can run parallely at a time
4) Faster execution; increases efficiency of system.
5) Cost Effective & ~~multi~~

* DisAdvantages.

1) Complex setup: Requires multiple processors (CPU) required ~~cost~~. time.

P.T.O

2) Expensive: many more processor cost higher.
3) Can create a Deadlock situation.
4) Maintaience is costly
5) Can create situation of process stops running should acknowldge to master node fi

b) List of Operating System Services:

1) Accounting
2) User Interface.
3) Error Detection
4) Program Execution
5) File System manipulation
6) Resource management.
7) I/O device allocation
8) Security & Protection.

1) User Interface
- User can interact with the system using user-interface service of operating system.
- User interact with Operating system in three ways or interf
1) Command-line Interface
2) Graphical user Interface
3) Batch Based Interface.

P. T. O

2) Program Execution

- Computer or operating system have smaller main-memory and larger Secondary-memory
- O.S can't access secondary memory programs they needs to bring in main-memory for execution.
- we not need to see allocation and deallocation O.S takes care of it.

3) File System manipulation.

- File system contains various directories & sub-directories which has file in it.
- ~~we~~ User manipulate the ~~ting~~ thing in File System or modification, updation, deletion etc......, comes under this services. Such as creating a file, deleting a file, Renaming a file etc.......

4) Resource managements.
- when process is in "running state" it requires several resources.
- Resources such as main memory, file, drivers etc.....

P.T.O

- Operating System does the job of allocating and deallocating resources.

5) Error Detection
- Error is an abnormal condit[ion] occurs and system stucks
- It's duty or job of operati[ng] system if any error is discovered then inform user via interface.
eg. mouse not working syntax error etc...

6) Security & Protection
- User can provide access to limited people in system is Security
- Protection is the files should not get corrupted etc...

04

c) Swapping

d) ~~Deadlock prevention~~

d) Deadlock Avoidance

Before the Deadlock occurs to avoid or to mitigate deadlock we require some algorithms.

Banker's Algorithm.

Banker algorithm is a algorithm where we can mitigate or avoid deadlock, like Bank (determine whether to give loan or not).

Steps 1: See given resources.

Step 2: Calculate need resources. ~~(Mutual Allocated =~~

(M - A) (mutual - Allocated)

Step 3: Given resources should be match with => ~~allo~~ needed.

Conditions

- Check which one less than available the add (allocated + available)
- Check which one less or equal and now available resources will be last iteration (allocated + ~~and~~ available).
- Determine and execute last one again and determine which can be executed & do what can execute and which can't. P.T.O

Step 4: Determine flow of process

Example

$P_4 \rightarrow P_1 \rightarrow P_2 \rightarrow P_3$ - - - - -

Step 5: Deadlock avoided successfully allocate likewise

Step 6: Stop

03

Ans.

c) Optimal

String / frame	7	0	1	2	0	3	0	4	2	3	0	3	2	1	2	0	1	7
f1	7	7	7	2	2	2	2	2	2	2	2	2	2	2	2	2	2	7
f2		0	0	0	0	0	0	4	4	4	0	0	0	0	0	0	0	0
f3			1	1	1	3	3	3	3	3	3	3	3	1	1	1	1	1
Result	F	F	F	F	H	F	H	F	H	H	F	H	H	F	H	H	H	F

H = Hit
F = fault

page fault = 09

P.T.O

LRU

String / Frame	7	0	1	2	0	3	0	4	2	3	0	3	2	1	2	0	1	7
f1	7	7	7	2	2	2	2	4	4	4	0	0	0	1	1	1	1	1
f2		0	0	0	0	0	0	0	0	3	3	3	3	3	3	0	0	0
f3			1	1	1	3	3	3	2	2	2	2	2	2	2	2	2	7
Result	F	F	F	F	H	F	H	F	F	F	F	H	H	F	H	F	H	F

H = Hit
F = fault

page fault = 12

Q)
2) i) ps
⇒ It list the process characteristics.

Syntax : $ps

Options	Syntax	description
-u	$ps -u ABC or $ps -u <username>	It list the processes related to specified user.
-a	$ps -a	It list the processes of all users
-e	$ps -e	list all the process with cmd user.
-l	$ps -l	list all commands with PPID, C etc.-

Example : $ps

PID	Time	TTYL	cmd
12345	00:00:00	pts/0	bash
12512	00:00:00	pts/0	bash

P.T.O

~~ii) wait~~

~~=> S/L stops the terminal for specific amount of time gi~~
~~It stops running of termin~~

~~Syntax: wait Time [options]~~

~~Syntax: wait Time [suffix]~~

~~Time should be specified in seconds.~~

~~Example: wait 5~~

~~It will stop terminal for 5 seconds.~~

~~iii)~~

ii) wait

=> If the process is in running stat it takes that process into waiting state by specifying its PID (Process Identification Number).

Syntax :- wait PID

Example :- wait 12345

P. T. O

iii) **Sleep**

It stops the terminal for specified time given by the user.

Syntax: $sleep time [suffix].

time should be specified in seconds.

Example

$sleep 15

It will stop terminal for 15 seconds

iv) **Kill**

It stops the execution of the running process.

It state that given is running of process it should running that process and exit / terminate and allowed for garbage collection of memory.

Syntax: $ kill pid

Example: $ kill 12345

Kills the process 12345.

There are also options available

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Last Written Page

PV-153

MAHARASHTRA STATE
BOARD OF TECHNICAL EDUCATION, MUMBAI 2223
(MAIN ANSWER BOOK)

Suppliment Sheet

To be filled by the Candidate

Subject :- OSY

Subject Code :- 2 2 5 1 6

Section :- I / II

Course and Year/Sem :- CO-5I

Date of Exam :- 0 3 / 1 2 / 2 0 2 4

Main Ans. Book	No. of Supplements	Total
1	01	02

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Sr. No. 27185193

Exam Seat No.: 2 4 8 8 2 4

Exam Center Code | Institute Code

Q. No.	a	b	c	d	e	f	g	h	i	j	k	l	m	n	Total	O.C.
1																
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Sign and Name of Examiner — Total

Q. No.	a	b	c	d	e	f	g	h	i	j	k	l	m	n	Total	O.C.
1																
2																
3																
4																
5																
6																

Sign and Name of Moderator — Total

Q6.

a) SJF

non-preemptive

GANTT chart

P2	P4	P3	P4

0   4   10   19   29

Completion time

$P_1 = 29$
$P_2 = 4$
$P_3 = 19$
$P_4 = 10$

Turnaround time
(Completion time - Arrival time)

$P_1 = 29 - 0 = 29$
$P_2 = 4 - 0 = 4$
$P_3 = 19 - 0 = 19$
$P_4 = 10 - 0 = 10$

Waiting time (TAT - BT)
(Turnaround time - Burst time)
$P_1 = 29 - 10 = 19$
$P_2 = 4 - 4 = 0$
$P_3 = 19 - 9 = 10$
$P_4 = 10 - 6 = 4$

P.T.O

Average waiting time

~~Total no. of~~

$$\frac{\text{All process waiting time}}{\text{Total no. of Process}}$$

$$\cancel{19+4+}\ \frac{P_1+P_2+P_3+P_4}{4}$$

$$= \frac{19+0+10+4}{4}$$

$$\boxed{\text{Avg waiting time} = 8.25 \text{ ms/unit}}$$

Round Robin (slice = 4ms)

GANTT chart

P1	P2	P3	P4	P1	P3	P4	P1	P3

0 4 8 12 16 20 24 26 28 29

Completion time

$P_1 = 28$
$P_2 = 8$
$P_3 = 29$
$P_4 = 26$

P.T.O

Turnaround time
(Completion time - Arrival time)

P1 = 28 - 0 = 28
P2 = 8 - 0 = 8
P3 = 29 - 0 = 29
P4 = 26 - 0 = 26

Waiting time

(Turnaround time - Burst time)

P1 = 28 - 10 = 18
P2 = 8 - 4 = 4
P3 = 29 - 9 = 20
P4 = 26 - 6 = 20

Average waiting time

$$\frac{P_1 + P_2 + P_3 + P_4}{4} = \frac{18+4+20+20}{4}$$

$= 15.5$ ms

Average waiting time = 15.5 ms/unit

06

c) Partitioning

When program or process from main memory is brought to main memory there are sections or block allocated to process, known as partitioning.
Each set of block holds one program.

Types of partitioning

1. Static/fixed partitioning
2. Variable/dynamic partitioning.

Variable partitioning

1) It is also known as dynamic partitioning.
2) As per requirement of the process size or program size blocks are allocated to in memory is variable partitioning.
3) They are dynamic allocation.
4) It's unlike fixed partitioning when process enters, size of block is determined on that process.

P.T.O

5) It is more flexible than the partitioning.

example

Process A (250MB) →

Process B (150MB)

Process C (500MB)

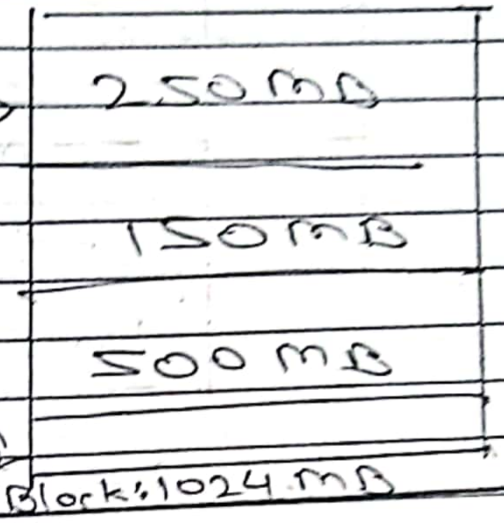


fig: 1.2.b

6) But, it faces problem of ~~internal~~ external fragmentation.

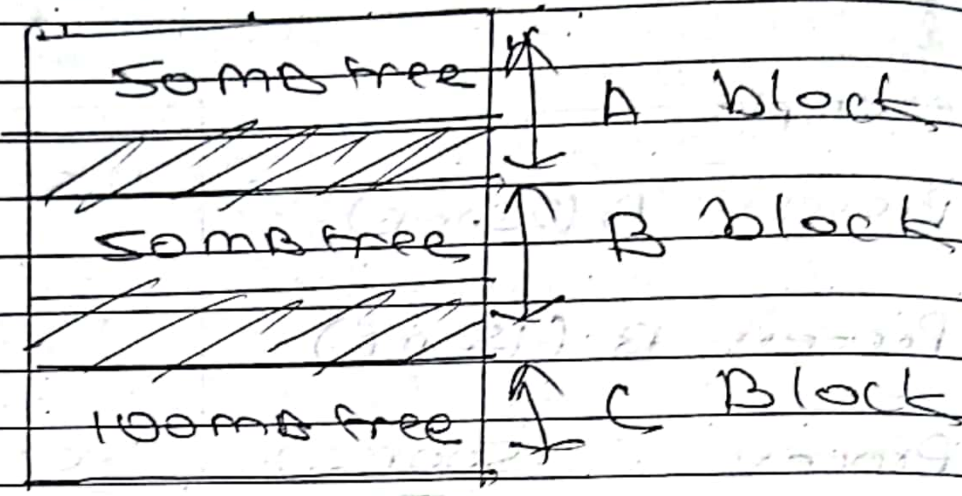
7) Even though there is space in memory but new process can't be allocated.

8) In above figure, 1.2.b if new process comes w/(600MB) and Process A & B release the memory then too due to scattered free space or holes can't allocate new process Because they are not in contiguous format.

9) It does not face problems of Internal fragmentation.

P.T.O.

External Fragmentation



If 600MB or 200MB process comes then to it will not allow.

Advantages

- Dynamic allocation.
- No internal fragmentation
- No wastage.
- slower but effective

06 Disadvantages

- Slower due to dynamic
- Internal fragmentation
- Costly.

Last Written Page

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